Random extras

COMP 4630 | Winter 2026

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Overview

- Object detection
- Autoencoders
- Generative adversarial networks (GANs)
- Diffusion models
- References and suggested reading:
  - Scikit-learn book: Chapter 17
  - Deep Learning book: Chapter 14, 20

Object Detection

R(egion)-CNN

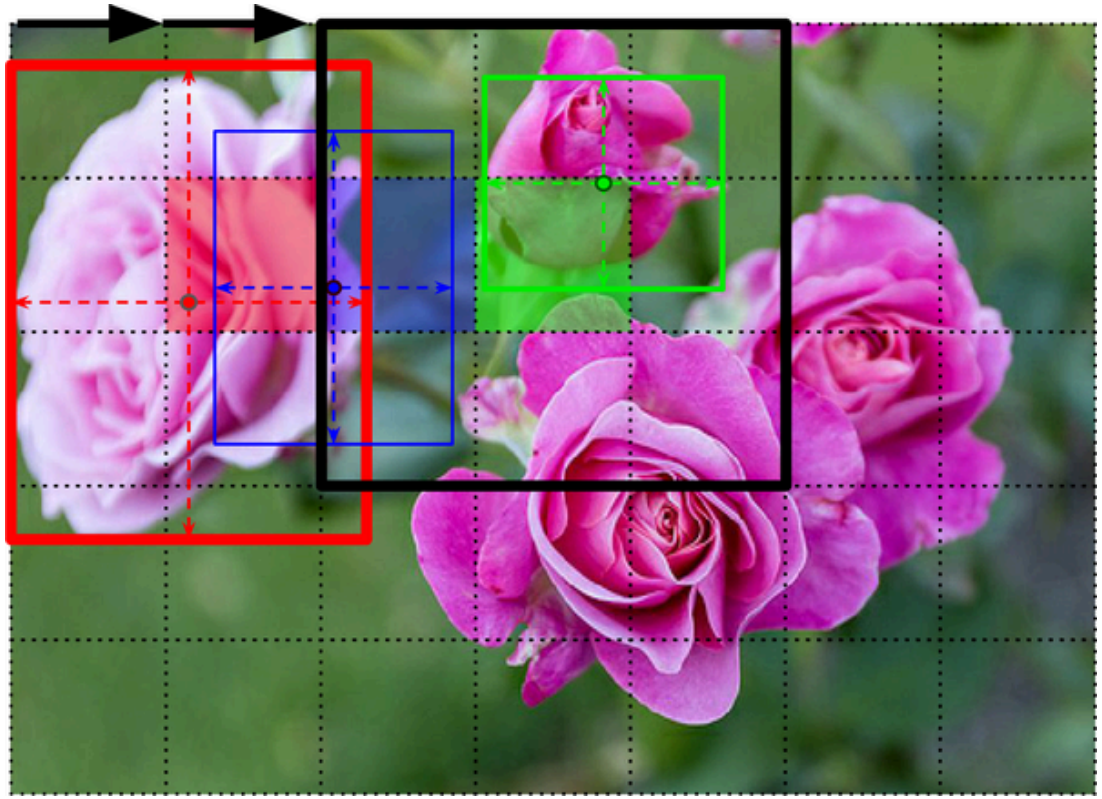

Fig 12-25 from Scikit-learn with Pytorch

You Only Look Once (YOLO)

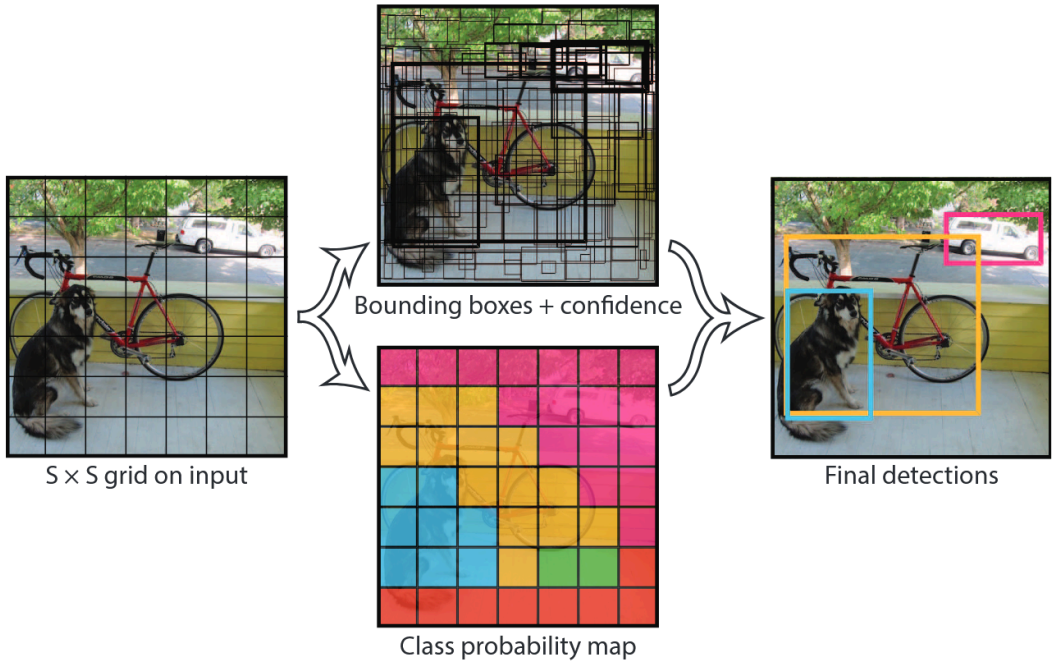

Fig 1 from Redmon et al

Autoencoders

- Like encoder-decoder networks, but the input and output are the same, minimizing:

$$\mathcal{L}(\mathbf{x}, f(g(\mathbf{x})))$$

  where f is the encoder and g is the decoder
- Autoencoders have been around since the 80s
- ❓ If we're just training a model to copy its input, what's the point?

What can we do with autoencoders?

- Autoencoders learn a **latent space** representation of the data, provided the latent space is **lower dimension** than the input (undercomplete)
- **Regularization** can also help the model learn a useful representation
- Once trained, the model can be used for:
  - Data compression
  - Information retrieval
  - Anomaly detection
  - Unsupervised pretraining

Unsupervised pretraining

- Rather than training (e.g.) a classifier from scratch, we can first train an **autoencoder**, then use the **encoder** part as the first half of the classifier

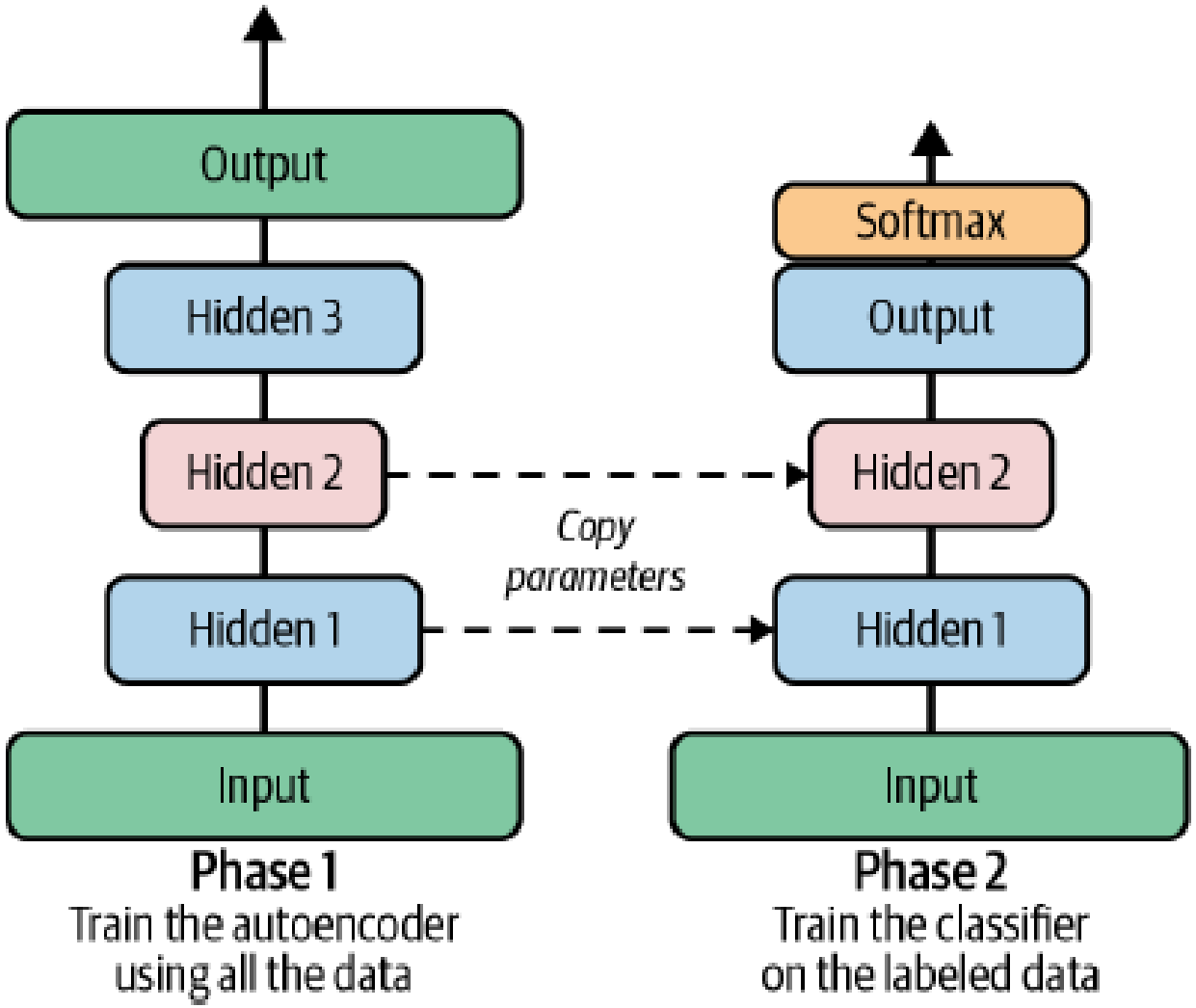

Figure 17-6 from the Scikit-learn book

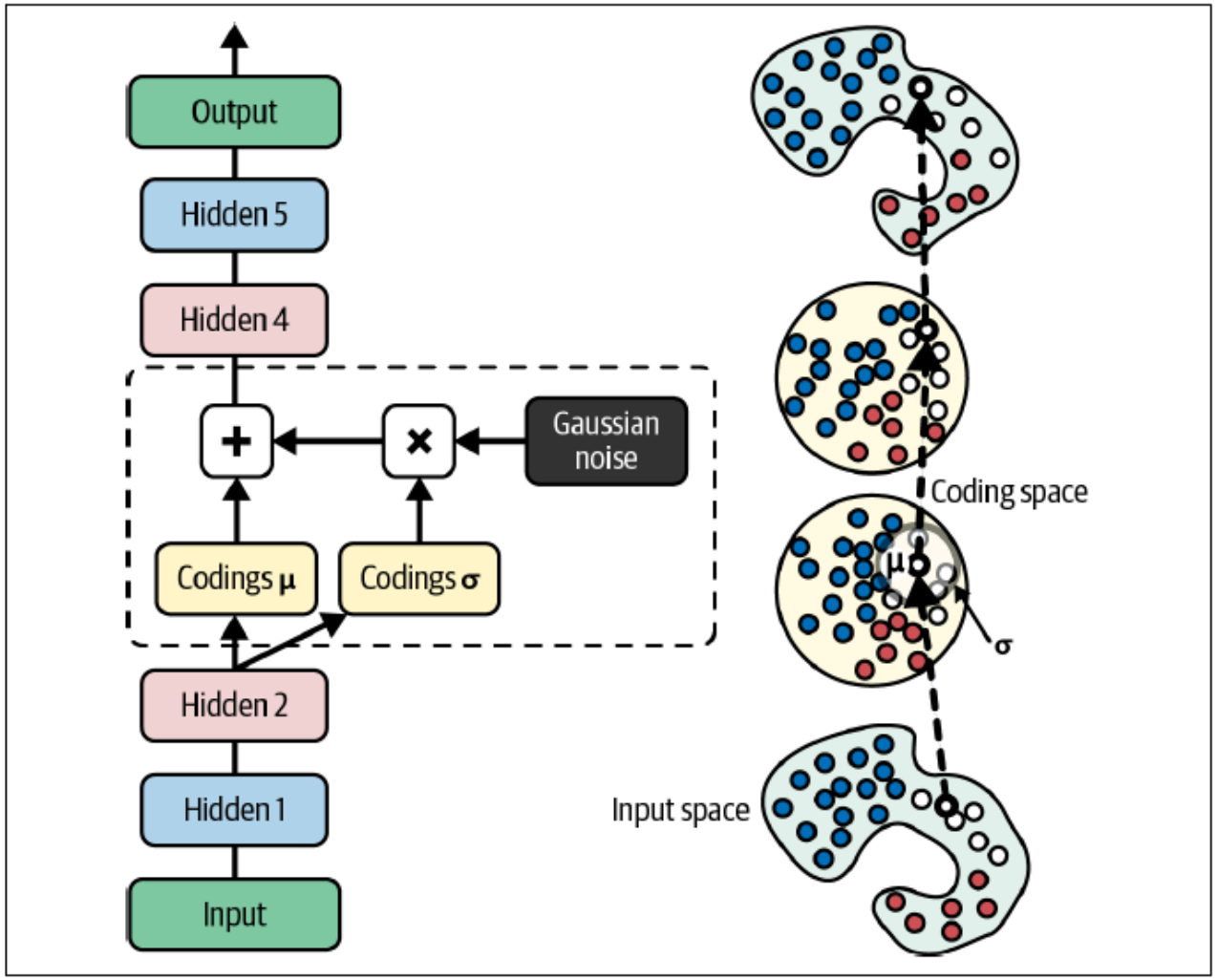

Figure 17-11. A variational autoencoder (left) and an instance going through it (right)

Variational Autoencoders

- Encoder produces a **distribution** (μ, σ)
- Decoder samples from $N(\mu, \sigma)$ to produce the output
- Loss includes a **KL divergence** term to push the distribution towards normal

Generative Adversarial Networks (GANs)

- VAEs can be used as generators (e.g. images), so how can we train them to generate what we actually want?
- In 2014, Ian Goodfellow et. al proposed training two networks simultaneously: a **generator** and a **discriminator**
  - The generator tries to produce realistic looking images
  - The discriminator is trained to classify images as real or fake
- Two-phase training loop:
  - Equal numbers of real and fake images used to train discriminator
  - Discriminator weights frozen while generator produces images with label of "real" - backpropagation updates only generator weights

GAN training challenges

- Training is a **zero-sum** game, where each "player" receives a payoff that the discriminator tries to maximize and the generator tries to minimize
- Equilibrium is reached when the discriminator performance is 50%, but **oscillations** can occur and convergence is not guaranteed
- **Mode collapse** happens when the generator figures out that it can produce a single image that fools the discriminator
- Still an active area of research!

> *"And since many factors affect these complex dynamics, GANs are very sensitive to the hyperparameters: you may have to spend a lot of effort fine-tuning them. In fact, that's why I used RMSProp rather than Nadam when compiling the models: when using Nadam, I ran into a severe mode collapse." - Aurélien Géron*

Diffusion Models

- First formalized in 2015 by Jascha Sohl-Dickstein et. al
- Not popular until 2020 when Jonathan Ho et. al introduced **DDPM**
- Concept: take an image and gradually add noise until it's unrecognizable, then train a model to reverse the process

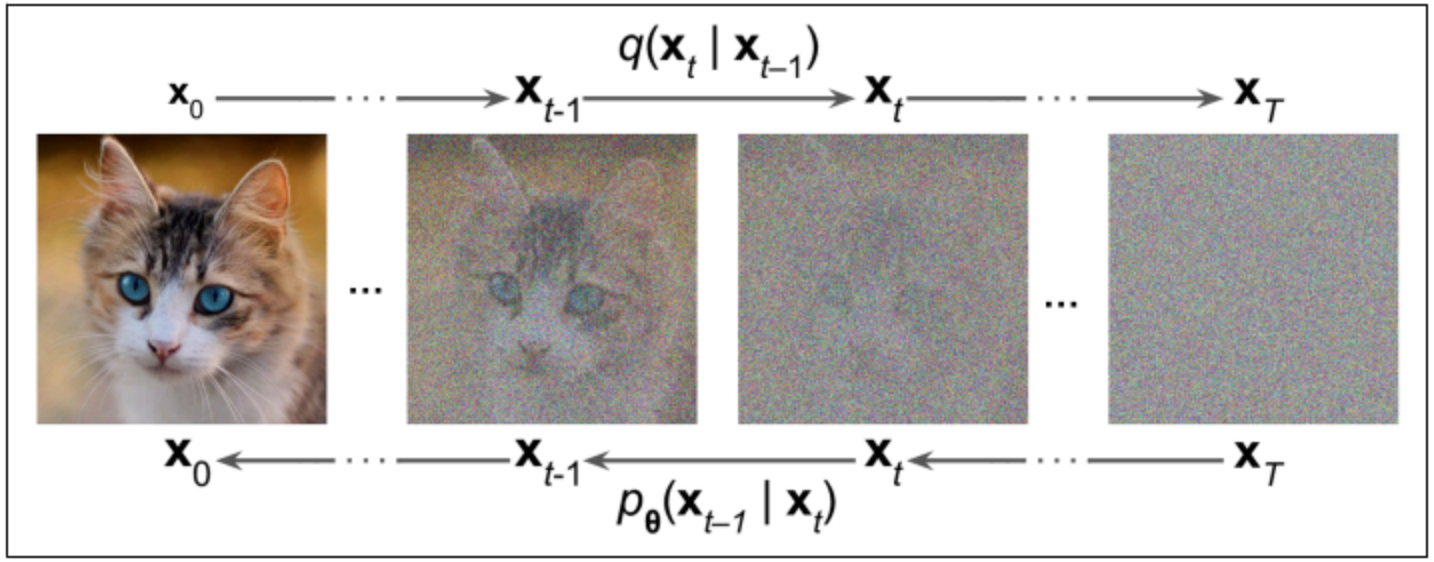

Figure 17-20. The forward process q and reverse process p

CLIP guidance

- DDPM, GANs, and VAEs are all capable of generating images given a **class label** that is part of the training data
- Image generation + NLP = magic
- Contrastive Language-Image Pretraining CLIP was trained on 400 million image-caption pairs to predict which caption goes with which image
- Final result is a general purpose model that can go both ways
- Actually open sourced by OpenAI!

That's all, next up is your project presentations!